Hazard Analysis and Risk Assessment (HARA)

Item: Autonomous Emergency Braking (AEB) System

Standard context: ISO 26262-3 concept phase HARA, structured for traceability, repeatability, and assessment evidence expected in Automotive SPICE / ASPICE environments. Automotive SPICE guidance explicitly emphasizes consistency and traceability, documented assessment processes, and objective evidence/work product management to support reproducible assessments. [vda-qmc.de]

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1) Document Control / Traceability Header

- Document ID: HARA-AEB-001
- Item Reference: AEB-ITM-001
- Item Revision Referenced: Rev A (based on previously defined AEB item definition)
- Analysis Level: Vehicle-level / Item-level hazardous event analysis
- Status: Preliminary concept-phase HARA
- Method: Malfunctioning behavior → Hazardous event (hazard + operational situation) → S/E/C classification → ASIL determination. The distinction between hazard, hazardous event, and operational situation is a core HARA principle in ISO 26262 concept-phase work. [waltercons...g-fusa.com], [en.wikipedia.org]

1.1 Scope of this HARA

This HARA covers malfunctioning behaviors of the AEB item in forward collision scenarios relevant to typical AEB functions, including lead vehicle stopped/moving/decelerating and, where included in item scope, pedestrian/cyclist scenarios. Public AEB references such as NHTSA NCAP and pedestrian AEB test material show these as representative AEB operating scenarios. [downloads....ations.gov], [iihs.org], [roadsafety-dss.eu]

1.2 Out of Scope

This document intentionally does not include:

- safety goals,
- functional safety requirements,
- technical safety mechanisms,

- architecture proposals,
- mitigations/design solutions.
  This keeps the work product aligned to the requested HARA-only scope.

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2) Rating Conventions Used

ISO 26262 HARA determines ASIL using Severity (S), Exposure (E), and Controllability (C). Public summaries of ISO 26262 consistently describe the rating classes as S0–S3, E0–E4, and C0–C3, with ASIL then derived from the standard decision matrix. [iso26262guide.com], [en.wikipedia.org]

2.1 Severity (S)

- S1: light to moderate injuries
- S2: severe injuries / life-threatening injuries with survival probable
- S3: life-threatening/fatal injuries possible
  These public explanations are consistent with widely used ISO 26262 training and guidance material. [motodemy.com], [iso26262guide.com]

2.2 Exposure (E)

- E1: very low probability
- E2: low probability
- E3: medium probability
- E4: high probability / frequent exposure
  Public ISO 26262 guidance describes E4 as frequent/common exposure in normal use and lower classes as progressively less common. [motodemy.com], [iso26262guide.com]

2.3 Controllability (C)

- C1: simply controllable
- C2: normally controllable
- C3: difficult to control or uncontrollable for the average driver/other road users
  Controllability is a distinct automotive factor in ISO 26262 and is especially important in ADAS functions where driver surprise and limited reaction time can strongly worsen control. [link.springer.com], [en.wikipedia.org]

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3) Operational Situations / Driving Scenario Catalog

The following scenario IDs are created to support bidirectional traceability from hazardous events back to the analyzed context, which is consistent with ISO 26262 HARA practice and ASPICE-style evidence expectations. [waltercons...g-fusa.com], [vda-qmc.de]

3.1 Scenario List

- OS-01 – Highway approach to stopped lead vehicle / traffic queue
  Host vehicle traveling at moderate-to-high speed in lane; stopped or near-stopped vehicle ahead. This maps well to public AEB "Lead Vehicle Stopped (LVS)" style scenarios. [downloads....ations.gov]
- OS-02 – Highway or expressway lead vehicle sudden deceleration
  Host vehicle follows lead vehicle that brakes hard; corresponds to "Lead Vehicle Decelerating (LVD)" type scenarios. [downloads....ations.gov]
- OS-03 – Urban/suburban approach to slower or stopped vehicle in same lane
  Lower speed, signalized traffic, queues, parked/stopped traffic transitions; conceptually aligned to common rear-end AEB use. [downloads....ations.gov]
- OS-04 – Urban pedestrian crossing / longitudinal pedestrian conflict
  Vehicle operates in city traffic where a pedestrian enters or remains in the host path; representative of pedestrian AEB assessments. [iihs.org], [roadsafety-dss.eu]
- OS-05 – Urban/suburban cyclist crossing or same-path cyclist conflict
  Cyclist intersects or occupies host path; representative of cyclist AEB scope in public literature. [roadsafety-dss.eu]
- OS-06 – Rural road approach to slower traffic / agricultural or utility vehicle
  Host vehicle approaches a slower same-lane target on a rural road with relatively high closing speed.
- OS-07 – Multi-lane road with adjacent-lane object, overhead structure, or non-relevant object in sensor field
  Relevant for wrong-target selection or false-positive braking.
- OS-08 – Driver already performing avoidance maneuver (steering around obstacle / lane change / cut-in handling)
  Rapid path relevance changes while the AEB item still evaluates collision likelihood.

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4) HARA Assumptions Used for this Preliminary Analysis

1. The analyzed AEB item includes forward collision intervention for same-lane vehicle targets, and includes pedestrian/cyclist targets only where the item scope says so. Public AEB testing frameworks distinguish vehicle scenarios from pedestrian/cyclist scenarios in exactly this way. [downloads....ations.gov], [iihs.org], [roadsafety-dss.eu]
2. The intended operational domain includes urban, suburban, rural, and highway forward-driving use, consistent with the earlier item definition and common AEB use contexts.
3. Exposure is judged from vehicle intended use, not from failure rate.
4. Controllability is judged for the average driver / typical surrounding traffic participant, not an expert driver. This aligns with public explanations of ISO 26262 controllability. [link.springer.com], [en.wikipedia.org]
5. Where multiple credible crash outcomes exist, the worst credible injury outcome in the analyzed scenario is used for Severity.

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5) HARA Worksheet (Traceable Hazard Log)

Legend:
MB = malfunctioning behavior ID
HE = hazardous event ID
OS = operational situation ID
ASIL derived using the ISO 26262 S/E/C risk matrix concept. [iso26262guide.com], [en.wikipedia.org]

5.1 Hazardous Events and ASIL Determination

HE ID	Source MB ID	Malfunctioning Behavior	Operational Situation / Scenario	Hazardous Event Description	Potential Harm	S	E	C	Resulting ASIL	Justification (traceable rationale)
HE-AE	MB-01	AEB fails to issue braking when a	OS-01	Vehicle continues at highway speed	High-energy rear-end crash involving	S3	E4	C3	ASIL D	S3: Highway rear-end into a stopped queue can credibly cause life-threatening/fatal injury.

HE ID	Source MB ID	Malfunctioning Behavior	Operational Situation / Scenario	Hazardous Event Description	Potential Harm	S	E	C	Resulting ASIL	Justification (traceable rationale)
B-01		collision with a stopped vehicle ahead is imminent		toward stopped queue / stopped lead vehicle without expected autonomous braking	host occupants and possibly secondary collisions					E4: Same-lane lead vehicle stopped scenarios are common in normal road use and are explicitly covered in public AEB programs as LVS-type scenarios. C3: Driver reaction time may be insufficient once the AEB intervention is absent at crash-imminent timing; avoidance margin is low. [downloads....ations.gov], [en.wikipedia.org], [link.springer.com]
HE-AEB-02	MB-02	AEB braking is delayed or insufficiently late for a rapidly decelerating lead vehicle	OS-02	Host vehicle fails to decelerate adequately when lead vehicle brakes hard in-lane	Rear-end crash at moderate-to-high relative speed	S3	E4	C3	ASIL D	S3: Sudden high-closing-speed rear-end crashes can produce severe to fatal injury. E4: Lead vehicle deceleration in traffic is a frequent road event; NHTSA publicly treats LVD as a representative AEB case. C3: Late realization of the malfunction leaves little controllability for an average driver. [downloads....ations.gov], [link.springer.com], [en.wikipedia.org]
HE-	MB-01 /	AEB fails to brake or	OS-03, OS-06	Host vehicle collides with	Injury to host occupants	S3	E4	C2	ASIL C	S3: Rural/suburban closing speeds can still

HE ID	Source MB ID	Malfunctioning Behavior	Operational Situation / Scenario	Hazardous Event Description	Potential Harm	S	E	C	Resulting ASIL	Justification (traceable rationale)
AEB-03	MB-02	brakes too late for slower/stationary same-lane vehicle at urban/suburban or rural speed		slower/stationary vehicle ahead in same lane	and possible injury to other vehicle occupants					make severe injury credible. E4: Slower/stopped same-lane traffic is common in intended use. C2: Compared with a fully stopped highway queue, the average driver may still retain some possibility to brake/steer if some margin remains, so controllability is judged slightly better than C3. [downloads....ations.gov], [link.springer.com], [en.wikipedia.org]
HE-AEB-04	MB-01	AEB fails to brake for a pedestrian in the host path	OS-04	Vehicle continues toward crossing / longitudinal pedestrian conflict without autonomous braking	Serious/fatal injury to pedestrian; possible host instability from late driver reaction	S3	E4	C3	ASIL D	S3: Pedestrian impact at urban traffic speeds can be life-threatening/fatal. E4: Urban crossings and pedestrian conflicts are frequent in intended city use and are explicitly represented in public pedestrian AEB protocols. C3: Pedestrians have very limited ability to influence outcome once conflict is imminent; driver controllability is also limited at short TTC. [iihs.org], [roadsafety-

HE ID	Source MB ID	Malfunctioning Behavior	Operational Situation / Scenario	Hazardous Event Description	Potential Harm	S	E	C	Resulting ASIL	Justification (traceable rationale)
										dss.eu], [en.wikipedia.org]
HE-AEB-05	MB-01	AEB fails to brake for cyclist in host path or same-path conflict	OS-05	Vehicle continues into cyclist conflict without autonomous braking	Serious/fatal injury to cyclist	S3	E3	C3	ASIL C *(preliminary)*	S3: Cyclist collision can credibly produce fatal or life-threatening injury. E3: Cyclist conflicts are regular in many urban/suburban environments but may be less universally frequent than same-lane vehicle scenarios across all markets. C3: Cyclist and driver typically have limited time/space to recover when the intervention is absent late. Public AEB literature treats cyclist protection as a recognized AEB use case. [roadsafety-dss.eu], [en.wikipedia.org], [link.springer.com]
HE-AEB-06	MB-03	AEB triggers emergency braking with no relevant obstacle	OS-01, OS-07	Sudden deceleration on highway or fast multilane road despite no collision threat	Rear-end impact from following traffic; loss of stability for host or following vehicles	S3	E4	C3	ASIL D	S3: Hard unexpected braking at speed can credibly result in severe/fatal crashes. E4: Highway and multilane driving are common intended-use conditions. C3: Surprise braking gives little time for the host driver and especially

HE ID	Source MB ID	Malfunctioning Behavior	Operational Situation / Scenario	Hazardous Event Description	Potential Harm	S	E	C	Resulting ASIL	Justification (traceable rationale)
										following traffic to respond; public AEB procedures even include false-positive assessment as an explicit concern. [downloads....ations.gov], [link.springer.com], [en.wikipedia.org]
HE-AEB-07	MB-04	AEB selects a non-relevant adjacent-lane / overhead / roadside object and brakes harshly	OS-07	Vehicle brakes for wrong target while actual host path is clear	Rear-end crash, side conflict during overtaking/merge, or loss of vehicle stability	S3	E4	C3	ASIL D	S3: At speed, wrong-target braking can trigger severe multi-vehicle outcomes. E4: Multi-lane roads and irrelevant objects in sensor field are common road conditions. C3: Driver surprise and surrounding-traffic dependence make recovery difficult in the available time. [downloads....ations.gov], [link.springer.com], [en.wikipedia.org]
HE-AEB-08	MB-05	AEB applies braking stronger than intended in low-speed urban scenario	OS-03	Host vehicle undergoes abrupt deceleration in traffic where only mild deceleration or no intervention	Rear-end from following vehicle, occupant injury due to unexpected deceleration	S2	E4	C2	ASIL B	S2: In lower-speed urban traffic, severe injury is possible but fatal outcome is less generally dominant than highway false braking. E4: Stop-and-go urban traffic is common. C2: At lower speeds, some

HE ID	Source MB ID	Malfunctioning Behavior	Operational Situation / Scenario	Hazardous Event Description	Potential Harm	S	E	C	Resulting ASIL	Justification (traceable rationale)
				would have been appropriate						surrounding drivers may still control the situation with prompt reaction, but not simply in all cases. [motodemy.com], [link.springer.com], [en.wikipedia.org]
HE-AEB-09	MB-06	AEB braking persists after collision threat is no longer relevant	OS-08	Driver begins steering around obstacle / cut-in clears / path no longer blocked, but AEB continues braking	Loss of directional control, secondary collision with adjacent traffic or roadside	S3	E3	C3	ASIL C *(preliminary)*	S3: A prolonged brake intervention during an avoidance maneuver can create a severe secondary crash. E3: Such dynamic path-relevance transitions occur regularly but less continuously than straight same-lane following. C3: Once the vehicle is already in a time-critical maneuver, additional unexpected braking is difficult for the average driver to manage. [link.springer.com], [en.wikipedia.org]
HE-AEB-10	MB-07	AEB unavailable / inactive due to internal malfunction while driver reasonably expects it to operate	OS-01, OS-02, OS-03, OS-04, OS-05	AEB does not intervene across intended supported scenarios although collision	Same harms as the corresponding “failure to brake” event for the active scenario	S3	E4	C3	ASIL D	Where the function is intended to operate and the internal malfunction suppresses intervention, the resulting hazardous event is equivalent to the relevant no-brake

HE ID	Source MB ID	Malfunctioning Behavior	Operational Situation / Scenario	Hazardous Event Description	Potential Harm	S	E	C	Resulting ASIL	Justification (traceable rationale)
				becomes imminent						case in the same scenario. Because AEB is assessed in those common forward collision contexts, the severity/exposure/controllability remain high. [downloads....ations.gov], [iihs.org], [roadsafety-dss.eu], [en.wikipedia.org]

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6) Consolidated Hazard List by Malfunctioning Behavior

This section helps assessment teams maintain one-to-many traceability between malfunction classes and hazardous events, which is useful for both ISO 26262 work products and ASPICE evidence structuring. [vda-qmc.de]

MB-01: No autonomous braking when braking is required

- Affects: HE-AEB-01, HE-AEB-03, HE-AEB-04, HE-AEB-05
- Typical consequence: failure to avoid/mitigate imminent forward collision

MB-02: Braking initiated too late / insufficiently late

- Affects: HE-AEB-02, HE-AEB-03
- Typical consequence: residual impact speed remains high

MB-03: Unintended braking without relevant target

- Affects: HE-AEB-06
- Typical consequence: false-positive emergency deceleration

MB-04: Wrong target selection / path relevance error

- Affects: HE-AEB-07

- Typical consequence: braking for non-threatening object

MB-05: Excessive braking intensity

- Affects: HE-AEB-08
- Typical consequence: unnecessary abrupt deceleration

MB-06: Brake release failure / braking persists after threat clears

- Affects: HE-AEB-09
- Typical consequence: destabilization during avoidance

MB-07: Function unavailable in supported use case

- Affects: HE-AEB-10
- Typical consequence: silent loss of intended intervention capability

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7) ASIL Summary

7.1 Highest ASILs Identified

The preliminary analysis indicates that the most critical AEB hazardous events are:

- Failure to brake for stopped or decelerating lead vehicle at speed
- Failure to brake for pedestrians
- Spurious harsh braking at highway speed / wrong-target braking

These produce ASIL D in this preliminary HARA because they combine:

- potentially S3 injury outcomes,
- common intended-use driving contexts (E4),
- and poor average-driver recoverability in crash-imminent or surprise-braking conditions (C3). [downloads….ations.gov], [iihs.org], [link.springer.com], [en.wikipedia.org]

7.2 Medium Criticality Events

Cyclist no-brake and prolonged-braking-during-avoidance are currently judged ASIL C (preliminary) because severity and controllability remain high, but exposure has been conservatively judged E3 pending market-specific usage data. This should be revisited with fleet data or regional ODD statistics. [roadsafety-dss.eu], [en.wikipedia.org]

7.3 Lower but Still Safety-Relevant Events

Excessive urban braking was judged ASIL B in this draft because the expected operating speeds and residual controllability are generally better than for highway false braking, although injury/crash risk still remains safety-relevant. [motodemy.com], [link.springer.com], [en.wikipedia.org]

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8) Traceability Notes for ISO 26262 and ASPICE CL3 Assessment Readiness

To support ISO 26262 traceability and ASPICE CL3-style assessment expectations, this HARA is structured so that each record contains:

- unique ID for item, malfunction, scenario, and hazardous event,
- a defined link from malfunction → scenario → hazardous event → S/E/C → ASIL,
- explicit rationale text for each classification,
- document-level assumptions that can be version-controlled,
- a reusable scenario catalog for future consistency, and
- a consolidated malfunction index to support downstream linkage.
  This aligns with public Automotive SPICE guidance on consistency/traceability, evidence, work product management, and documented assessment processes. [vda-qmc.de]

Recommended traceability fields (for your formal work product)

If you later put this into Word or Excel, add the following columns/metadata fields to make it audit-friendly:

- Item ID
- Requirement/Feature Reference
- Operational Situation ID
- Hazardous Event ID
- Malfunction ID
- Assumptions Reference

- Reviewer
- Review Date
- Rationale Version
- Status (Draft/Baselined/Approved)
- Change History Ref

That keeps the artifact suitable for review, baselining, and change-impact analysis in a CL3 process environment. [vda-qmc.de]

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9) Open Points / Review Items

The following items should be explicitly reviewed in the project HARA workshop before baselining:

1. Whether pedestrian and cyclist coverage are inside or outside the actual item scope. Public AEB references often separate vehicle, pedestrian, and cyclist evaluations. [downloads....ations.gov], [iihs.org], [roadsafety-dss.eu]
2. Whether the intended ODD justifies E3 vs E4 for cyclist and complex urban scenarios.
3. Whether "AEB unavailable while expected" should be split by scenario rather than treated as an umbrella malfunction.
4. Whether regional road-use data warrants changing exposure classes for highway, rural, or urban use.
5. Whether towing, low-friction, trailer, steep-gradient, or other special operating contexts require additional hazardous events.